DANY HAIKAL BIN YAHYA

Bachelor of Computer and Communication System Engineering with Honours, UPM

- Muar, 84200 - +6016-4672855 - danyhaikal79@gmail.com - www.linkedin.com/in/dany-haikal-44a27817b/

Summary

Results-driven Computer and Communication System Engineering (Hons.) final year student from Universiti Putra Malaysia with hands-on exposure Artificial Intelligence, Autonomous Robot, Embedded System Design, Microprocessor Technology, Computer Aided Engineering Drawing, Electronic Communication Circuits, Communication System Design and Simulation, Programming, Network Troubleshoot and Network Topology Design and Simulation, and hardware configuration through internship and academic research through my bachelor. Adept at translating analytical insights into actionable engineering improvements that enhance performance, safety and cost efficiency. Looking for anygraduate trainee or permanent engineering position that can be started on 1st September 2026.

Education

Universiti Putra Malaysia - Bachelor of Computer and Communication Systems Engineering with Honours — Selangor, Malaysia
CGPA: 3.43 / 4.000 - Second Upper Class Honour — **Oct 2022 - Sep 2026**

Centre of Foundation Studies Universiti Putra Malaysia – Foundation in Agricultural Science — Selangor, Malaysia
CGPA: 3.55/4.00 – First Class Honour — **Aug 2021 – Jun 2022**

Project Involvement

Intelligent Poultry & Feed monitoring System to Detect Feed Level and Predict Optimize Feed Volume

- Designed the data collection setup before development started using Jetson Nano, Pi Camera, and Several Iot Sensors to read the temperature, humidity, CO2, and ammonia data.
- Collect and annotate data before feeding it to AI for training, validation and testing.
- Trained and fine tuned the AI model hyperparameter and tested it using sample data to optimize AI accuracy without overfitting.
- Validate and evaluate the AI model performances

IoT - CO2 and MoldRisk Prediction System Using AI

- Deployed and configured an IoT monitoring system using environmental sensors to collect environmental data
- Developed IoT data pipelines to process, store, and manage real-time sensor data efficiently.
- Processed sensor data using Python and KNN-based data imputation techniques, and trained an LSTM AI model for 30-minute prediction of MoldRisk and CO2 levels.
- Integrated Flask API with the system to automate prediction data transfer and visualize real-time readings and AI predictions on an interactive dashboard.

RFID Smart Inventory Management System

- Led the system design process, focusing on hardware selection, data flow architecture, and user interface optimization to ensure efficient device functionality and usability.
- Designed and developed a warehouse RFID detection system using ESP32, PN532 RFID reader, and LCD touchscreen display for real-time inventory tracking.
- Developed and simulated the system backend and frontend to manage data operations including sorting, adding, deleting, and processing scanned records.

Stray Dog Detection System using Artificial Intelligence

- Led project planning and task distribution to ensure an efficient and organized AI system development process.
- Collected, preprocessed, and managed datasets from multiple platforms for AI model training and validation.
- Trained, and evaluated a YOLOv12 AI model for real-time stray dog detection based on project objectives and research analysis.
- Developed a Streamlit dashboard integrated with map visualization and alert notifications for real-time stray dog detection monitoring.

Experiences

Beyond2U Sdn. Bhd. **Selangor, Malaysia**

System Engineer Intern Jul 2026 - Oct 2026

- Execute tech refresh replacement following provided guidelines on several AmBank branches, while troubleshooting any expected or unexpected problems.
- Setup new ip and domain configurations to ensure the network is private, so that no external party can access the data inside the AmBank domain.
- Transferred all the required data from the old pc to the new one without missing any single file.
- Installed BitLocker on the PC so no unrecognized device can connect to the PC through wired connection.
- Executed KillDisk on the old PC to hard disk to make sure no possible way the data can be leaked.
- Deal with AmBank Branch Manager and user for arrangement on PC deployment to avoid unproductive and inefficient work flow.

TECHNICAL SKILLS

- **Programming Languages :** Python, Java, C++, HTML, Verilog
- **Engineering & Design Software :** MATLAB, Simulink, AutoCAD, SolidWorks, Microsoft Visio, Microsoft Project
- **Networking & Communication Tools :** GNS3, OptiSystem, MQTT, Oracle VirtualBox
- **FPGA & Embedded Systems :** Intel Quartus Prime
- **Internet of Things (IoT) & Data Visualization :** Node-RED, Grafana, Streamlit
- **Database Management :** MySQL, InfluxDB
- **Artificial Intelligence & Machine Learning :** YOLO, LSTM, Neural Networks, Google Colab, Ollama
- **Productivity Tools :** Microsoft Excel, Microsoft Word, Microsoft PowerPoint

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Leadership Experiences

Universiti Putra Malaysia **Selangor, Malaysia**

Participants in Build Your Own Machine Learning Workshop 2024

- Participated in a hand-on "Build Your Own Machine Learning" workshop, gaining practical experience in building, training, and evaluating machine learning models

Universiti Putra Malaysia **Selangor, Malaysia**

Wire Bond Course Introduction & Career Talk, organized by NXP Malaysia 2024

- Participated in a specialized course on wire bonding technology, learning about key processes, materials, and quality considerations in semiconductor manufacturing

Taman Negara Malaysia **Pahang, Malaysia**

CSR Project: Unity Through STEM 2024

- Led my team in introducing ferrofluid to the native people and give them exposure about STEM

Universiti Putra Malaysia **Selangor, Malaysia**

CSR Project: Rimba ke Menara Gading 2023

- Led Bateq native people in exploring urban side of Malaysia at Kuala Lumpur while ensuring their safety

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References

Prof. Dr. Mohd Adzir bin Mahdi
Lecturer/Academic Advisor
Universiti Putra Malaysia
Tel: +603-9769 6438
Email: mam@upm.edu.my